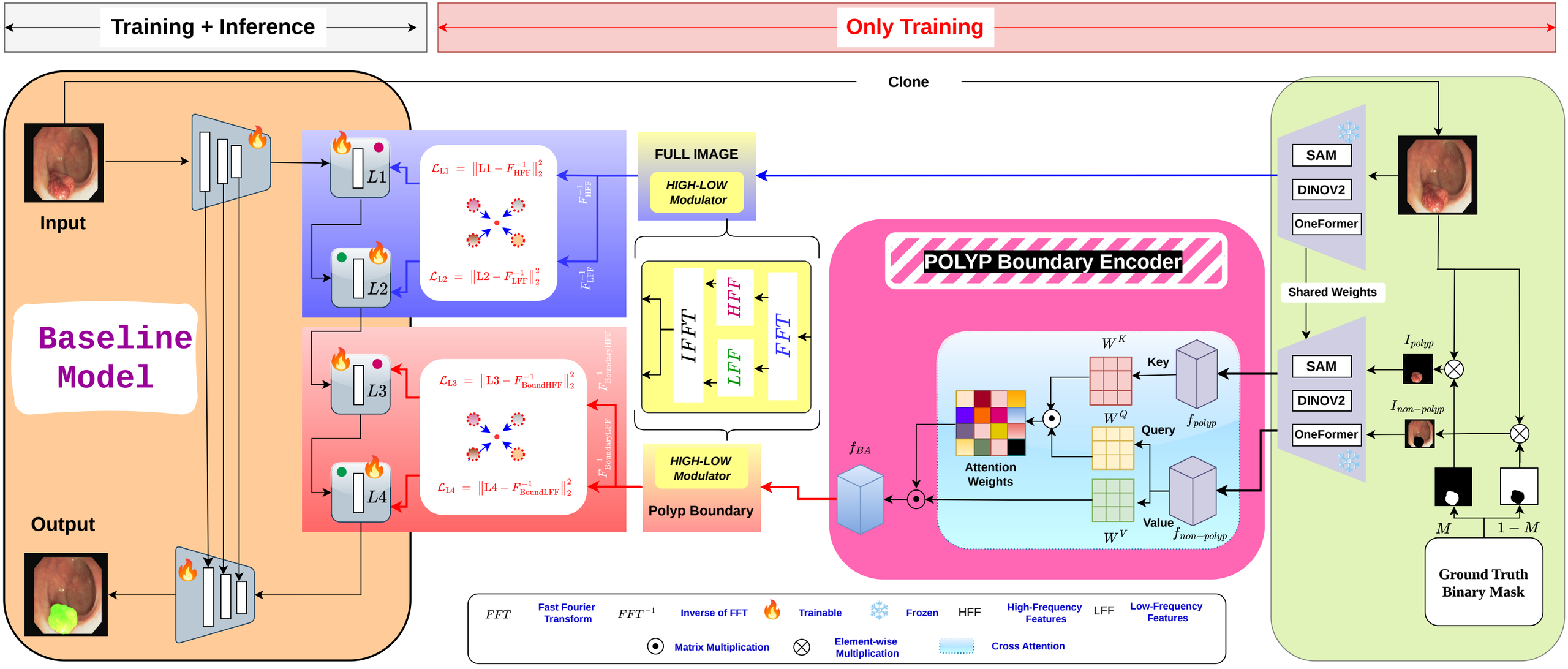
Training + Inference
Only Training
Clone
Input
Output
Baseline Model
L1
L2
L3
L4
$\mathcal{L}_{L1} = \|L1 - F^{-1}_{HFF}\|_2^2$
$\mathcal{L}_{L2} = \|L2 - F^{-1}_{LFF}\|_2^2$
$\mathcal{L}_{L3} = \|L3 - F^{-1}_{BoundHFF}\|_2^2$
$\mathcal{L}_{L4} = \|L4 - F^{-1}_{BoundLFF}\|_2^2$
F^{-1}_{HFF}
F^{-1}_{LFF}
$F^{-1}_{BoundaryHFF}$
$F^{-1}_{BoundaryLFF}$
FULL IMAGE
HIGH-LOW Modulator
IFFT
HFF
LFF
FFT
HIGH-LOW Modulator
Polyp Boundary
POLYP Boundary Encoder
f_{BA}
W^K
Key
W^Q
Query
W^V
Value
f_{polyp}
$f_{non-polyp}$
Attention Weights
SAM
DINOV2
OneFormer
Shared Weights
SAM
DINOV2
OneFormer
I_{polyp}
$I_{non-polyp}$
M
$1 - M$
Ground Truth Binary Mask
FFT Fast Fourier Transform
FFT^{-1} Inverse of FFT
Trainable
Frozen
HFF High-Frequency Features
LFF Low-Frequency Features
Matrix Multiplication
Element-wise Multiplication
Cross Attention